

What Are Recursive Functions and Stack Overflows?

A *recursive function* is a function that calls itself. This *shortest.py* program is the shortest possible example of a recursive function:

Python

```
def shortest():  
    shortest()  
  
shortest()
```

The preceding program is equivalent to this *shortest.html* program:

JavaScript

```
<script type="text/javascript">  
function shortest() {  
    shortest();  
}  
  
shortest();  
</script>
```

The `shortest()` function does nothing but call the `shortest()` function. When this happens, it calls the `shortest()` function again, and that will call `shortest()`, and so on, seemingly forever. It is similar to the mythological idea that the crust of the Earth rests on the back of a giant space turtle, which rests on the back of another turtle. Beneath that turtle: another turtle. And so on, forever.

But this “turtles all the way down” theory doesn’t do a good job of explaining cosmology, nor recursive functions. Since the call stack uses the computer’s finite memory, this program cannot continue forever, the way an infinite loop does. The only thing this program does is crash and display an error message.